

IGNN: An Inference Graph Neural Network for Data Sensitivity Assessment Considering Data Inference Leakage in Cyber-Physical Power Grid

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Abstract—Modern Cyber-Physical Power Grids (CPPG) rely heavily on fine-grained data exchange, yet they face the concealed threat of Data Inference Leakage (DIL), where attackers infer sensitive parameters from seemingly non-sensitive data. Existing risk assessment methods typically focus on direct cyber-attacks or employ static binary classifications, failing to capture the dynamic risk propagation caused by complex physical and statistical data couplings. To address this, this paper proposes a unified data sensitivity assessment framework based on a novel Inference Graph Neural Network (IGNN). We first construct an inference relation graph that integrates both model-driven and data-driven correlations. An analytic hierarchy process (AHP) model is constructed to calibrate the intrinsic sensitivity baseline. Subsequently, a novel inference-weighted Graph Attention Network (GAT) is developed to quantify the final sensitivity by aggregating the risk propagating through the inference pathways. Experiments on IEEE 30-, 57-bus, ACTIV 200-bus, and case 1354-pegase systems demonstrate that the proposed method effectively quantifies data sensitivity and identifies high-risk data susceptible to DIL, exhibiting strong robustness across different system scales. Quantitative results indicate that the high-sensitivity data identified by our method yield a highly threatening inference accuracy of up to 0.9145, significantly outperforming GCN-based and rule-based benchmarks.

Index Terms—risk assessment, data security, cyber-physical power grid, graph neural network.

I. NOMENCLATURE

BUS	Bus identifier (ID)
P_d	Active power demand at the bus
Q_d	Reactive power demand at the bus
P_i	Active power injection at the bus
Q_i	Reactive power injection at the bus
G_s	Shunt conductance
B_s	Shunt susceptance
V_m	Voltage magnitude
V_a	Voltage angle
$baseKV$	Base voltage (kV)
V_{max}	Upper limit of voltage magnitude
V_{min}	Lower limit of voltage magnitude

LAM_p	Locational Marginal Price (LMP) for active power
LAM_q	Locational Marginal Price (LMP) for reactive power
MU_{vx}	Kuhn-Tucker multiplier for voltage upper limit constraint
MU_{vi}	Kuhn-Tucker multiplier for voltage lower limit constraint
P_g	Generator active power output
Q_g	Generator reactive power output
Q_{max}	Maximum generator reactive power
Q_{min}	Minimum generator reactive power
P_{max}	Maximum generator active power
$fBUS$	From-end bus identifier
$tBUS$	To-end bus identifier
r	Branch series resistance
x	Branch series reactance
b	Total branch shunt susceptance
$rateA$	Long-term thermal rating
$rateB$	Short-term thermal rating
$rateC$	Emergency thermal rating
$status_{bran}$	Branch status (1=in-service, 0=out-of-service)
Ang_{min}	Minimum voltage angle difference
Ang_{max}	Maximum voltage angle difference
A	Grid topology adjacency matrix
B	Bus admittance matrix
P_f	Active power flow at the from-end
Q_f	Reactive power flow at the from-end
P_t	Active power flow at the to-end
Q_t	Reactive power flow at the to-end
P_s	Active power line loss
Q_s	Reactive power line loss
CI	Consistency Index of the AHP model
CR	Consistency Ratio of the AHP model
$\mathcal{G}$	Inference relation graph model
$\mathcal{V}$	Set of graph nodes corresponding to data fields
$\mathcal{E}$	Set of graph edges corresponding to inference relations
W	Weight matrix of the graph edges
Cor_{pc}	Pearson correlation coefficient
Cor_d	Distance correlation coefficient
Cor_{cos}	Cosine similarity coefficient
R	Inference matrix representing the structure and strength of relations
α_{ij}	Attention coefficient indicating the importance

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	of node j to i
K	Number of attention heads in the multi-head attention mechanism
M	Mask matrix used to block unreasonable inference paths
$\hat{s}_i$	Final predicted sensitivity of data field i
$\mathcal{L}$	Loss function for IGNN training

II. INTRODUCTION

With the development of advanced metering technology and distributed energy resources, power grids have been transformed into highly interconnected cyber-physical systems (CPSs) that rely on continuous fine-grained data exchange for real-time generation, transmission and trading. Both the volume and granularity of data sharing and disclosure have increased dramatically in cyber-physical power grids (CPPGs), and abundant sensitive information such as key parameters or privacy has been included in CPPG data. Therefore, it is crucial to assess the risk of sensitive information leakage to ensure the security of data sharing or disclosing in CPPG.

The existing studies of risk assessment for CPPG target external threats to system stability and mainly focus on evaluating the severity and probability of cyber attacks (e.g., false data injection or denial-of-service) and analyzing system vulnerabilities [1]. Nevertheless, the CPPG data faces more complicated risks of sensitive information leakage. While traditional cyber attacks compromise sensitive data through direct breaches (e.g., bypassing firewalls), a more concealed threat: *Data Inference Leakage* (DIL), arises and has been verified in multiple application scenarios in CPPG [2], [3]. DIL means attackers utilize accessible non-sensitive data to infer protected sensitive data by exploiting data correlations, without directly breaching the database, which may unconsciously bypass the protections against cyber attacks.

Generally, sensitive data include critical system parameters (e.g., topology) and state variables, the disclosure of which could cause significant losses, while non-sensitive data usually comprises public market information or general measurements. Since DIL enables security risk to propagate from sensitive to non-sensitive data, the simple labeling of sensitive and non-sensitive data in CPPG are insufficient. It is necessary to incorporate the impact of risk propagation caused by DIL and quantify the risk of CPPG data as numerical metric. This metric reflects the level of security concern required by the data, so it is defined as *data sensitivity*. Data sensitivity assessment can serve as an important part for effective risk assessment by quantifying the data-related security risks, complementing the traditional risk assessment which focuses on physical and network levels of the system.

The main challenges of data sensitivity assessment under DIL threats include, but not limited to: i) The complexity of inference mechanisms hinders accurate risk identification. Unlike direct attacks, DIL relies on heterogeneous relationships, including model-driven physical laws (e.g., power flow equations) and data-driven statistical correlations. Existing methods often fail to integrate these diverse coupling relationships into a unified view. ii) Lack of a framework to quantify risk

propagation. In the context of DIL, the sensitivity of one type of data depends not only on its information content but also on its inferential connectivity to other sensitive data. Current regulations and rule-based methods typically employ sensitive and non-sensitive classification, which cannot capture the continuous magnitude of sensitivity propagating through the inference network. This deficiency results in an inability to comprehensively evaluate the risk level of CPPG data, especially by considering the impact of DILs.

To address the above challenges, we propose a data sensitivity assessment framework under the threat of DIL in CPPG based on a novel Inference Graph Neural Network (IGNN). The main contributions are summarized as follows.

First, a unified framework for the quantitative assessment of data sensitivity in CPPG is established to address the DIL threat. To capture the risk propagation path, we construct an inference relation graph where data fields act as nodes and both model-driven and data-driven correlations serve as edges. Within this framework, an Analytic Hierarchy Process (AHP) is employed to initialize the intrinsic sensitivity of nodes, serving as the baseline for subsequent propagation analysis.

Second, a novel IGNN method is proposed to quantify the final sensitivity by aggregating risks propagated through the inference relation graph. Recognizing that inference leakage is inherently directional and causal, we design an inference-weighted Graph Attention Network (GAT) to perform node embedding on the mixed interaction graph. This approach enables the dynamic calculation of a sensitivity metric for each data field, reflecting both its inherent security level and the inference risk derived from DIL.

Third, the proposed data sensitivity assessment method is tested on IEEE 30-, 57-bus systems, the Illinois ACTIV 200-bus system, and the Case 1354-pegase system. The experiments show that our method is effective in quantitatively assessing data sensitivity and identifying the risk data that implied potential DIL risks in CPPG, with promising robustness and stability.

The remainder of this paper is organized as follows. Section III presents related work on risk assessments in CPPG. Section IV introduces problem statements and preliminaries. Section V develops the data sensitivity assessment method and provides a detailed description of the inference association graph model and the IGNN method. Section VI presents numerical experiments and analysis of results. Section VII concludes the paper with a discussion of future work.

III. RELATED WORK

With the deepening informatization of power systems, CPPG faces a widening spectrum of cyber and data security risks. However, traditional risk assessment frameworks focus primarily on attacks on communication networks and electronic devices, paying limited attention to data leakage. To bridge these gaps, recent studies have introduced sensitive data identification and data inference methods, thus enriching the knowledge of data-centric security risks. These researches will be discussed in detail in three subsections.

A. Risk Assessment for CPPG

Risk assessment in Cyber-Physical Power Grids (CPPG) has evolved from static vulnerability analysis to dynamic, data-driven modeling. Existing methodologies can be broadly categorized into probability-based approaches, AI-driven detection, and graph-learning frameworks. Early research primarily focused on identifying vulnerabilities within SCADA systems [1] and modeling coordinated attacks across cyber-physical layers [4], [5]. To quantify these risks, probabilistic methods have been widely adopted. For instance, Yuan et al. [6] combined Bow-tie diagrams with Bayesian networks to model causal relationships in industrial process security, while Ten et al. [7] established frameworks for modeling attack and defense mechanisms. Although these methods provide interpretability for known attack paths [8], they heavily rely on expert-defined rules and static topologies, making them insufficient for capturing the complex, implicit data-driven risks.

To address the complexity of modern grids, recent studies have leveraged advanced technologies for real-time security management. Tang et al. [9] proposed a Digital Twin-driven approach to enhance grid resilience, enabling real-time simulation of cyber risks. Similarly, AI-driven detection techniques [10], [11] have been introduced to automate threat identification using deep learning. While these methods significantly improve the speed of risk identification, they fundamentally differ from our objective. As noted in [12], these frameworks predominantly focus on system-level consequences (e.g., physical outages, equipment damage) rather than data-level security. They lack fine-grained metrics to evaluate the potential risks of individual data fields, leaving the system vulnerable to indirect information leakage threats.

Graph learning has emerged as a promising tool for capturing topological dependencies in risk assessment. Notably, Sun et al. [13] proposed a graph embedding-based approach to automatically assess system risks and mitigate threats. However, while they utilize graph embedding, its objective is to identify vulnerable nodes to prevent physical system failures (e.g., cascading blackouts). It models risk propagation based on physical connectivity (transmission lines) but ignores the correlations between heterogeneous data assets. In contrast, our proposed framework tries to construct a specific inference relation graph that captures both model-driven and data-driven couplings. This allows us to quantify the risk of indirect data leakage, a dimension often overlooked by traditional methods. A comparative summary of these approaches and the proposed method is presented in Table I.

B. Sensitive Data Identification

The most common method of assessing data sensitivity is to identify data that contain sensitive information. The early approaches for the identification of sensitive data were largely based on dictionary-based and rule-based matching techniques guided by published criteria. Although feasible for specific datasets, these methods are highly dependent on the completeness and timeliness of the dictionaries and rules, which limits their adaptability.

With increasing data complexity and scale, machine learning (ML) and deep learning (DL) methods have gained prominence in sensitive data identification [14]–[17]. Ma et al. [18] introduced a deep content-based method leveraging natural language processing for industrial power data. Convolutional Neural Networks (CNNs), enhanced by pre-trained models and attention mechanisms, have also been explored for the identification of sensitive data in power systems [19]. Recurrent Neural Networks (RNNs), such as LSTMs, have shown utility in sequence-based data such as financial records [14]. Transformer-based models such as BERT have demonstrated superior performance in natural language processing for sensitive data detection [15]. However, most intelligent methods focus on detecting explicitly sensitive data using direct content or labels and often lack the ability to systematically find data that have correlated risks with the other sensitive data.

C. Data Inference

A vital goal of our method is to explicate the inference relations between the CPPG data and to assess the impacts. Existing studies of inference modeling can be roughly divided into data-driven and model-driven approaches. Data-driven inferences are mainly investigated in machine learning models, which includes membership inference, model inversion, data reconstruction, etc [20]–[22]. The idea can be concluded as inferring sensitive information, such as membership or training data, from the output of the machine learning model [23]. These studies reveal some data inference relations but are hard to utilize in CPPG scenarios due to the gap of definitions of sensitive information in actual physical systems.

Model-driven inference is generally realized by adopting known physical models to establish mathematical relations between general data and high-sensitivity data. Therefore, these studies focus mainly on data scenarios with rich physical properties (such as CPPG). For instance, Kekatos et al. [3] were the first to propose a grid topology (which is widely recognized as highly sensitive data) inference method using non-sensitive electricity prices based on the economic dispatch model. Some others also utilize the abundant available measurements to infer critical operating states and parameters [24], such as voltage phasor, topology [25], emergency power capacity [26], generator parameters [27], etc. Although these studies can provide a comprehensive knowledge of data inference relations, it still remains unaddressed to establish a complete mapping of inference relations between CPPG data, since the existing efforts mostly focus on some single relation between a few types of data.

IV. PROBLEM STATEMENT AND PRELIMINARIES

A. DIL Threat and Data Sensitivity Assessment

As we mentioned in Section II, DIL is a more intractable threat of data leakage than direct data theft or breach. Although sensitive data are usually isolated from the public environment during storage and transmission processes, they are still highly correlated with non-sensitive data due to the sharing of consistent underlying mechanisms. Thus, associations among multiple data fields may provide a certain inference path

TABLE I
COMPARISON OF PROPOSED METHOD WITH STATE-OF-THE-ART RISK ASSESSMENT APPROACHES

Category	Representative Works	Core Methodology	Assessment Focus	Limitation regarding DIL
Probability-based	[6], [7]	Bayesian Networks, Bow-tie Diagrams	Causal analysis of system faults and attacks	Relies on static rules, cannot capture implicit data correlations or inference paths.
AI & Digital Twin	[9], [10]	Digital Twin, Deep Learning Detection	Real-time resilience and intrusion detection	Focuses on system outages or continuity, ignores the sensitivity levels of specific data fields.
Graph-based	[13]	Graph Embedding	Node vulnerability and attack mitigation	Models physical topology propagation, not data-level risk correlation analysis.
Proposed Method	This Work	Inference Graph Neural Network (IGNN)	Data security management	Designed for capturing the DIL risks to comprehensively assess the data-level risks.

that links sensitive and non-sensitive data, giving rise to a new inference-based data leakage. Real-world events have demonstrated that risk propagation via data inference is a tangible threat. For instance, in the broader security context, the *Strava Heatmap* incident revealed that non-sensitive fitness tracking data shared by soldiers could be used to infer the secret locations and layouts of military bases [28].

Especially in CPPGs, with the development of sensor-rich smart grids and electricity markets, more seemingly non-sensitive data are collected and given to the public or authorized participants for the system-control and economic-dispatch demands, and abundant physical constraints also provide feasible DIL paths for potential attackers. A prominent example occurred during the 2015 Ukraine power grid cyber attack [29]. Before the blackout, attackers conducted long-term reconnaissance to harvest network credentials and basic system logs. While these individual data points were not the final target, they allowed attackers to infer the critical SCADA topology and operational logic required to manually trip circuit breakers, causing the first large-scale power grid paralysis induced by a cyber attack. Similarly, in the 2021 Iran nuclear facility leak, adversaries combined commercial construction contracts, engineering plans, and satellite imagery to precisely infer the progress of nuclear projects and identify physical vulnerabilities (e.g., blind spots or weak structural points) [30]. These events indicate that without assessing the risk propagation inherent in data inference before data sharing, aggregated low-sensitivity data may expose the system's most critical information.

Therefore, DIL threats are prominent and highlighted in our data sensitivity assessment method. To systematically analyze the aforementioned threats, we formally define the mechanism of DIL and the mathematical formulation of the data sensitivity assessment problem in CPPGs.

Let the set of all data fields in a CPPG be denoted as $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$. $\mathcal{V}$ can be partitioned into two subsets: the confidential dataset $\mathcal{V}_C$ and the general dataset $\mathcal{V}_G$. The mechanism of DIL relies on the existence of inference models derived from physical laws or statistical correlations. The inference models can be defined as a set of mapping functions $\mathcal{F}$. Then, based on subset of general data $\mathbf{x} \subseteq \mathcal{V}_G$ and domain knowledge $\mathcal{K}$ (e.g., power flow equations or historical databases), DIL can be presented as an inference function

$f \in \mathcal{F}$ to estimate target confidential data $y \in \mathcal{V}_C$:

$$\hat{y} = f(\mathbf{x} \mid \mathcal{K}) \quad (1)$$

where $\hat{y}$ is the inferred value of y .

Thus, even if $\mathbf{x}$ consists solely of general data, it implies a high security risk due to its capability to expose y . Consequently, the risk propagates from the target y to the source $\mathbf{x}$ through the inference pathway f .

On this basis, the core problem of data sensitivity assessment is to quantify the risk level of each data field while considering the impact of potential DIL threats. The problem can be defined as finding a mapping function Φ that assigns a sensitivity score s_i to each data field. The sensitivity score s_i is determined by two components: i) The intrinsic sensitivity s_i^{int} of v_i based on its static attributes (e.g., impact on confidentiality, integrity, and availability), typically evaluated by existing regulations or standards. ii) The propagated sensitivity s_i^{prop} derived from DIL. Mathematically, the data sensitivity assessment problem can be formulated as solving for the equilibrium state of sensitivity distribution on a risk propagation graph. The objective is to calculate the final sensitivity $\mathbf{s} = [s_1, s_2, \dots, s_n]$:

$$s_i = \Phi \left(s_i^{int}, \sum_{v_j \in \mathcal{N}(v_i)} w_{ji} \cdot s_j \right) \quad (2)$$

where $\mathcal{N}(v_i)$ is the set of neighbors connected to v_i in the risk propagation graph, w_{ji} represents the weight of risk propagation from v_i to v_j , and $\Phi(\cdot)$ is a nonlinear aggregation function.

Solving this problem requires identifying the complex structure of the risk propagation graph and designing an effective algorithm for the aggregation function $\Phi(\cdot)$, which motivates the proposed IGNN-based framework.

B. Graph Attention Network

The graph attention network (GAT) uses a self-attention mechanism to assign learnable weights to different nodes in a neighborhood. A GAT layer updates each node's features by weighting and aggregating information from its neighbors. For a node i with features $\mathbf{h}_i$ and neighbors $j \in \mathcal{N}(i)$, the layer first applies a shared linear transformation $\mathbf{W}$ to the

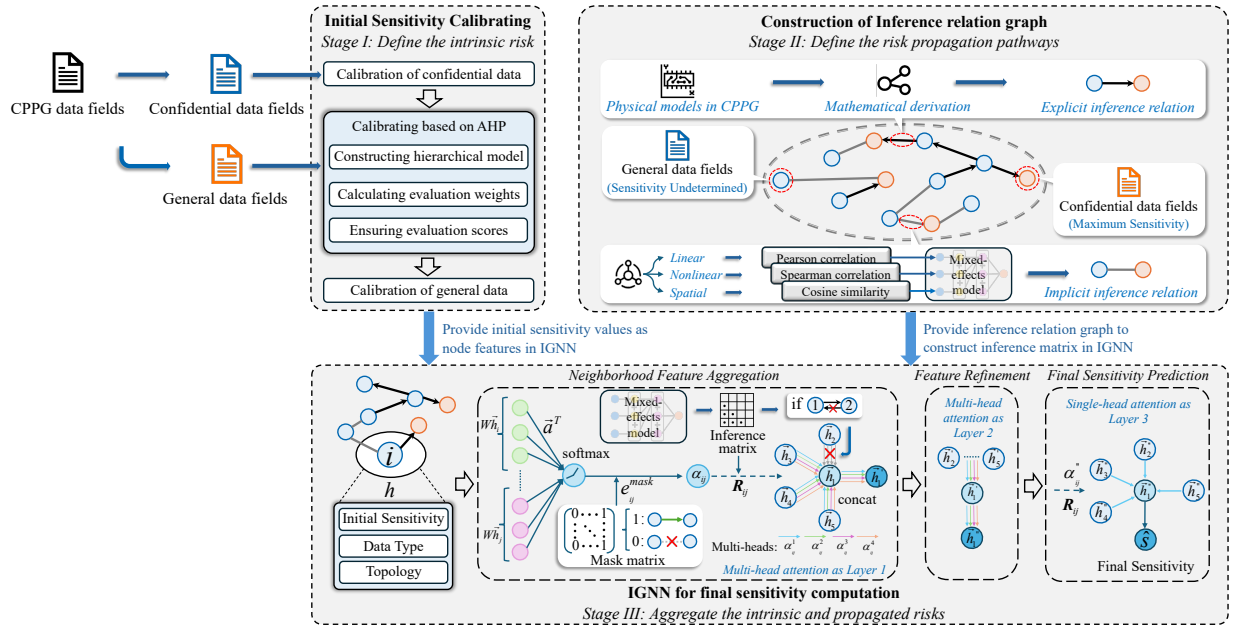


Fig. 1. The framework of data sensitivity assessment based on IGNN. The framework consists of three stages: Stage I establishes the baseline sensitivity evaluation based on intrinsic data risks. Stage II constructs the inference graph to map potential risk propagation pathways by DIL. Stage III utilizes the initial sensitivity as node features and employs IGNN to aggregate the propagated risks by inference graph, updating the final sensitivity values.

node features. An attention coefficient α_{ij} is then calculated for each neighbor, quantifying its importance to node i :

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_k]))} \quad (3)$$

where $\mathbf{a}$ is a learnable weight vector of a feedforward network. The final output feature $\mathbf{h}'_i$ for the node is a weighted sum of its neighbors' features, passed through a non-linear activation function σ :

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j \right) \quad (4)$$

Additionally, GATs often use multi-head attention, where independent attention mechanisms are computed in parallel and their features are aggregated.

V. DATA SENSITIVITY ASSESSMENT UNDER DIL THREAT IN CPPG

A. Initial Sensitivity Calibrating

As shown in Fig. 1, the first step of our method is to establish a baseline estimate of the sensitivity for the target data fields, preparing for the assessment of risk propagation by DIL threats. The calibration of the initial sensitivity is accomplished by analyzing the security risks inherent to each data field and evaluating their severity according to the relevant standards and regulations for power system data security.

1) *Confidential Data Calibrating*: Some critical data (such as line parameters and nodal states) are commonly considered sensitive and cannot be disclosed outside the grid. These data are referred to as *confidential data* and therefore can be directly given the maximum sensitivity. Confidential data

are considered the “seed” of sensitive information to provide sources of data leakage risks in other data.

Confidential data are calibrated referring to two regulations: The Basic Rules for Electricity Market Information Disclosure [31] issued by the China National Energy Administration, and the PJM Electricity Market Information Disclosure Mechanism [32] developed by PJM in the United States. Based on the above prior knowledge, 13 confidential data fields are labeled (see the calibration in Fig 2). All other data are treated as *general data* that need further evaluation of the initial sensitivity. The range of sensitivity is set to $[0, 1]$, and the sensitivity of confidential data is 1.

2) *General Data Calibrating based on AHP*: Unlike confidential data, general data carry no obvious sensitive elements and are usually treated as non-sensitive by existing methods. However, beyond direct exposure of sensitive information, other potential risks of general data should also be considered in assessing data sensitivity, especially in data sharing scenarios. Therefore, multiple risk dimensions are extracted from existing efforts to construct an evaluation method based on the Analytic Hierarchy Process (AHP) to score the initial sensitivity of general data.

The AHP-based evaluation is a systematic decision-making method that combines qualitative and quantitative analysis. The basic principle involves three steps. First, decompose the complex decision problem into a hierarchical structure (e.g., goal, criteria, and alternatives). Second, construct a judgment matrix by performing pairwise comparisons of elements at each level using a standardized scale. Third, calculate the weight vector and verify the logical rationality using a Consistency Ratio (CR) check.

Referring to the existing classification of CPPG data security risks [33], six key dimensions of the security risk

TABLE II
EVALUATION DIMENSIONS OF THE AHP MODEL

No.	Dimension	Detailed description
a	Leakage Risk	The leakage of data may lead to malicious attacks, commercial disadvantages, or other security issues.
b	Access Control	The strictness of access control: unauthorized access could trigger severe security problems.
c	Data Tampering Impact	The tampering of data may lead to erroneous operations or system failures, potentially causing security accidents.
d	Data Consistency	The inconsistencies of data may cause deviations in grid scheduling and operations.
e	System Dependency	System has high dependency on the data which directly affects the normal operation of the power grid.
f	Data Loss Impact	Data loss can result in operational faults, business interruptions, or decision-making errors within the power grid.

assessment are extracted to build the hierarchy of our AHP method. Detailed evaluation dimensions are presented in Table II.

Then a judgment matrix is constructed to quantify the relative importance of each evaluation dimension in Table III. Each element a_{ij} in the matrix represents the importance of the i -th dimension relative to the j -th dimension based on Saaty's 1-9 scale (e.g., scale 1 indicates equally important, while scale 9 indicates extremely more important).

TABLE III
JUDGMENT MATRIX OF THE EVALUATION DIMENSIONS

	a	b	c	d	e	f
a	1	3	5	7	9	2
b	0.333	1	3	5	7	0.5
c	0.2	0.333	1	3	5	0.333
d	0.143	0.2	0.333	1	3	0.2
e	0.111	0.143	0.2	0.333	1	0.143
f	0.5	2	3	5	7	1

The specific values in Table III were determined by synthesizing expert knowledge from power system security operations and analyzing historical data security incidents in transmission grids. For example, *Leakage Risk* is assigned a higher weight relative to *System Dependency* (i.e. $a_{ae} = 9$) because key parameter leakage is often irreversible and can lead to global attacks, whereas system dependency issues are often localized and recoverable.

It is also worth noting that the AHP framework is scenario-adaptive. While the matrix in Table III reflects the consideration of a standard power grid scenario (emphasizing confidentiality and integrity), the method can be applied to other systems (e.g., distributed energy systems or electricity markets) by adjusting the pairwise comparison scores. For example, in a real-time market scenario, *Data Tampering* or *Data Loss* might be assigned higher importance than *Leakage Risk* to prioritize business continuity. Users can update the judgment matrix based on their specific operational requirements of the target system.

Furthermore, regarding the subjectivity of expert scoring, it is important to recognize that risk assessment in critical infrastructure inevitably requires the incorporation of expert prior knowledge to capture qualitative security demands that purely data-driven methods may overlook. The AHP method standardizes this process. More importantly, in our proposed framework, the AHP scores serve only as the initial assessment. The final sensitivity is predominantly determined by the risk propagation through the objective inference relation graph constructed from physical models. The aggregation

mechanism of the IGNN acts as a calibration process to adjust the minor fluctuations in the initial AHP weight evaluation. Therefore, the final IGNN output can to some extent avoid over-dependence on the initial subjective assumptions.

Then, the weight of each evaluation dimension can be calculated based on the judgment matrix:

$$w_i = \frac{1}{n} \sum_{j=1}^n a'_{ij} \quad (5)$$

where $a'_{ij} = \frac{a_{ij}}{\sum_{i=1}^n a_{ij}}$ is the element of the normalized judgment matrix and n is the number of evaluation dimensions. The final weights are presented in Table IV.

TABLE IV
WEIGHTS OF THE EVALUATION DIMENSIONS

Dimension	Weight
Leakage Risk	0.3956
Access Control	0.1846
Data Tampering Impact	0.0990
Data Consistency	0.0513
System Dependency	0.0278
Data Loss Impact	0.2417

In addition, consistency checks are demonstrated to ensure that the judgment matrix meets the consistency requirements of the AHP. By computing the eigenvalues of the judgment matrix, the maximum eigenvalue $\lambda_{\max}$ can be obtained. Then, the Consistency Index (CI) and CR can be calculated as follows:

$$CI = \frac{\lambda_{\max} - n}{n - 1}, CR = \frac{CI}{RI} \quad (6)$$

where RI is the Random Consistency Index corresponding to the order of the matrix n . Based on AHP [34], the consistency check is passed if $CR < 0.1$. In our method, the consistency check yielded $\lambda_{\max} = 6.261$, $CI = 0.052$, and $CR = 0.042$ (with $RI = 1.25$). Since $CR < 0.1$, the judgment matrix satisfies the consistency requirement and the weights of the evaluation dimensions are proved to be reliable.

Then, for any general data field, its initial sensitivity can be calculated by:

$$s = \sum_{i=1}^6 w_i \cdot s^i \quad (7)$$

where $w_1, w_2, \dots, w_6$ are the weights of the evaluation dimensions, and $s^1, s^2, \dots, s^6$ are the scores of each dimension given by the estimation of the risk level on each dimension. The scoring utilizes a scale from 1 to 9, where 1 indicates *very insensitive* and 9 indicates *very sensitive*. In the section

of experiments, we will present and use our own evaluated risk scores of target data fields to calibrate the initial sensitivity.

In summary, the AHP method is employed solely to quantify the initial sensitivity ($s = s^{int}$) of general data fields based on their operational importance (e.g., impact on system stability or data consistency). For instance, a data field like power demand (P_d) may receive a high AHP score due to its critical role in dispatch decisions (dimension “e” in Table II). s^{int} serves as the initial node feature input to the IGNN model. Subsequently, the IGNN aggregates the risk propagated from connected confidential data fields to calculate the final sensitivity ($\hat{s} = s^{final}$). Therefore, the assessed sensitivity values are not direct outputs of the AHP scoring. They are the composite result where the AHP score provides the baseline operational value, and the IGNN superimposes the dynamic risks derived from the DIL pathways.

B. Construction of Inference Relation Graph Model

1) *Definition of Inference Relation Graph:* To assess data sensitivity under the impact of DIL threats in CPPG, the propagation of data security risks is represented by a graph model $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, \mathbf{W} \rangle$, which converts the inference relations to edges between graph nodes. The detailed definitions of the inference relation graph are as follows.

- Node set $\mathcal{V}$: Each node $v_i \in \mathcal{V}$ ($i \in [1, N]$) corresponds to a specific data field, where N is the number of target data fields. The attributes of each node include not only its initial sensitivity (see Section V-A for details), but also the data type and topological features (see the specific design in Section V-C1).
- Edge set $\mathcal{E}$: Each edge $e_k = \langle v_i, v_j, d \rangle \in \mathcal{E}$ ($k \in [1, M]; i, j \in [1, N]$) represents an inference relation between the data fields v_i and v_j , where M is the number of inference relations. The relations include two types: directed ($d = 1$) and undirected ($d = 0$), which indicate that the inference can only be achieved from v_i to v_j or can be achieved by bi-directions, respectively. The construction of inference relations will be detailed in Section V-B2 and Section V-B3.
- Weight matrix $\mathbf{W}$: The weight w_{ij} quantifies the strength of the inference relation between v_i and v_j , which reflects the confidence level of a DIL path. The numerical values of the weights can be derived from the data-driven correlation metrics described in Section V-B3.

2) *Construction of Explicit Inference Relations Based on Physical Models:* Based on physical models and constraints, the confidential data may be directly computed or indirectly estimated using equations or optimization approaches from other general data. Therefore, the identified model-driven relations are considered as explicit inference relations in CPPG.

Depending on the characteristics, model-based inference relations can be classified into two categories: strong and weak inference relations. A strong inference relation indicates that one data field can be calculated or estimated based on the other data field (or with knowledge of the model, such as parameters). For example, the bus voltage magnitude and phase angle can be estimated using active and reactive

injection power based on the state estimation model [24]. In addition, the grid topology or line parameters can be calculated using injection power based on the inverse power flow model [25]. A weak inference relation refers to cases where there is no direct physical derivation between the data fields but certain constraints or influences exist. For example, the actual voltage magnitude is constrained by the operational range of the bus voltage. Moreover, the impact of generator status on generator output power, which, although not directly computable through derivation, affects the feasible range of power values.

Based on the prior knowledge of physical models in CPPG [35], the explicit inference relations can be extracted from the known data relations, which has been shown in Table V and Table VI. Considering the direction of risk propagation and the inherent computational irreversibility, we use two primary criteria to ensure the correctness of the causal directionality of inference relations: i) When an inference relation bridges confidential and general data, the security risk naturally propagates from the sensitive source to the non-sensitive target. Thus, the relation is directed from confidential to general data. ii) When the calculation principle is mathematically one-way, the relation follows that direction.

As detailed in Table V, the identified strong inference relations primarily connect confidential to general data, adhering to the first criterion, while weak inference relations function as unidirectional constraints (e.g., a physical limit constrains a variable's value, but the value does not determine the limit). Based on these reasons, we define all explicit inference relations as unidirectional and represent them with directed edges in the graph (shown in Fig. 2).

TABLE V
STRONG INFERENCE RELATIONS BASED ON PHYSICAL MODELS

Data Relation	Underlying Mechanism
$\langle P_i, V_m \rangle$ $\langle P_i, V_a \rangle$ $\langle Q_i, V_m \rangle$ $\langle Q_i, V_a \rangle$	AC Power Flow Model [24]
$\langle fBUS, A \rangle$ $\langle tBUS, A \rangle$	Bus Admittance Model [35]
$\langle P_i, V_a \rangle$ $\langle P_i, A \rangle$ $\langle P_i, x \rangle$	DC Power Flow Model [36]
$\langle LAM_p, B \rangle$	Economic Dispatch Model [3]

3) *Construction of Implicit Inference Relations Based on Data-Driven Correlations:* In contrast to model-driven relations, data-driven correlations can also be utilized to infer sensitive data. These relations are defined as *implicit inference relations* because they represent latent statistical dependencies that lack a direct, analytical physical expression. Regarding the directionality, unlike the computational irreversibility often found in explicit relations, statistical correlation is inherently symmetric (i.e., the correlation coefficient). Consequently, an adversary can exploit this mutual dependence to perform inference in either direction. To reflect this mathematically symmetric mechanism of risk propagation, implicit inference

TABLE VI
WEAK INFERENCE RELATIONS BASED ON PHYSICAL CONSTRAINTS

Data Relation	Underlying Mechanism
$\langle V_{max}, V_m \rangle$	Voltage upper limit
$\langle V_{min}, V_m \rangle$	Voltage lower limit
$\langle P_{max}, P_g \rangle$	Active generation power capacity
$\langle Q_{max}, Q_g \rangle$	Reactive generation power upper limit
$\langle Q_{min}, Q_g \rangle$	Reactive generation power lower limit
$\langle rateA, P_f \rangle$	Thermal stability limits
$\langle Ang_{max}, V_a \rangle$	Static stability limits
$\langle LAM_p, P_f \rangle$	LMP-Congestion correlation
$\langle MU_{vx}, V_m \rangle$	KKT complementary conditions
$\langle MU_{vi}, V_m \rangle$	KKT complementary conditions
$\langle LAM_g, Q_g \rangle$	Reactive marginal cost correlation
$\langle status_{bran}, P_f \rangle$	Influence of branch status on power flow

relations are modeled as undirected edges in the inference relation graph.

Three metrics are adopted to characterize implicit inference relations from the perspective of linear, nonlinear, and spatial correlations. First, the *Pearson correlation coefficient* (PCC) is adopted to quantify the linear correlation between the CPPG data. Considering the issue that CPPG data (such as bus data, branch data, and generator data) may have different dimensions, making direct PCC computation infeasible, Principal Component Analysis (PCA) is used to map different data fields into the same dimensionality. Assuming two data fields $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t]$ and $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_t]$ which include t snapshots of the data values, use PCA to reduce the dimensionality of the high-dimensional field to match the dimensionality of the low-dimensional field. Then the PCC [37] can be computed as follows.

$$Cor_{pcc} = \frac{\sum_{i=1}^t (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{y}_i - \bar{\mathbf{y}})}{\sqrt{\sum_{i=1}^t \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2} \sqrt{\sum_{i=1}^t \|\mathbf{y}_i - \bar{\mathbf{y}}\|^2}} \quad (8)$$

Then, considering that CPPG data may exhibit nonlinear relations, we introduce the *Distance Correlation* (dCor) in case correlations are not strictly linear. Since dCor can be computed between variables of different dimensionality, we calculate dCor [38] directly as follows without PCA:

$$dCov^2(\mathbf{X}, \mathbf{Y}) = \frac{1}{t^2} \sum_{i=1}^t \sum_{j=1}^t \mathbf{A}_{ij} \mathbf{B}_{ij} \quad (9)$$

$$Cor_d = \frac{dCov(\mathbf{X}, \mathbf{Y})}{\sqrt{dCov(\mathbf{X}, \mathbf{X}) dCov(\mathbf{Y}, \mathbf{Y})}} \quad (10)$$

where $\mathbf{A}_{ij} = a_{ij} - \bar{a}_{i.} - \bar{a}_{.j} + \bar{a}_{..}$, $a_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$, then $\mathbf{B}_{ij}$ can be obtained similarly.

Moreover, because most CPPG data are highly topology-related (such as measurements on branches or bus nodes), *Cosine Similarity* is also employed to measure spatial con-

sistency between data fields. After the same PCA processing on $\mathbf{X}$ and $\mathbf{Y}$, the cosine similarity [39] can be calculated as:

$$Cor_{cos} = \frac{\sum_{i=1}^t \langle \mathbf{x}_i, \mathbf{y}_i \rangle}{\sqrt{\sum_{i=1}^t \|\mathbf{x}_i\|_2^2} \sqrt{\sum_{i=1}^t \|\mathbf{y}_i\|_2^2}} \quad (11)$$

After the computation of the correlation metrics between CPPG data, we establish a linear mixed-effects model (LMM) [40] to fuse the multiple metrics to obtain a weight coefficient, for quantitatively characterizing the strength of inference relations. The LMM model is formulated as follows:

$$r = \mu + \mathbf{c}\boldsymbol{\omega} + S + \epsilon \quad (12)$$

where,

- r represents the inference strength between two data fields and corresponds to the weight of an edge in the inference relation graph, which means $r_{ij} = w_{ij}$ between data v_i and v_j .
- μ is the baseline correlation between two data fields.
- $\mathbf{c} = [Cor_{pcc}, Cor_d, Cor_{cos}]$ is the correlation vector.
- $\boldsymbol{\omega} = [\omega_1, \omega_2, \dots]$ is the fixed-effect vector that quantifies the contribution of each correlation measure.
- S represents the random effects component and ϵ is the noise term.

S and ϵ follow normal distributions with zero mean and unknown variance, which can be presented as: $S \sim \mathcal{N}(0, \sigma_s^2)$, $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$

LMM can assess the importance of different correlation measures and determine the final correlation metric R by estimating the parameters $\mu, \boldsymbol{\omega}, \sigma_s, \sigma_\epsilon$. To estimate the parameters, a training dataset containing the correlation vector $\mathbf{c}$ and the known inference strength values $\hat{r}$ is first obtained. $\mathbf{c}$ can be calculated based on the snapshots of data field values. And, $\hat{r}$ can be obtained by the pre-defined strength of the explicit inference relations. Specifically, as we mentioned in Section V-B2, inference relations are classified into strong and weak relations and should be assigned with different strength values. To ensure metric consistency between the model-driven physical relations and the data-driven statistical correlations within our unified framework, we align the strength settings with widely accepted statistical benchmarks for the Pearson correlation coefficient [41]. In standard statistical interpretation, a correlation coefficient $|r| \geq 0.7$ typically signifies a “strong” relationship, while $|r| \leq 0.3$ indicates a “weak” or negligible relationship. Adopting these standard thresholds allows us to map the deterministic nature of strong inference relations (equality constraints) to the high-confidence value of 0.7, and the loose coupling of weak inference relations (inequality constraints) to the low-confidence value of 0.3. This alignment provides a standardized scale to quantify heterogeneous inference mechanisms in the LMM. The loss function of LMM is defined as follows:

$$\mathcal{L}_{lmm} = \min_{\mu, \boldsymbol{\omega}, \sigma_s, \sigma_\epsilon} \sum_{i=1}^{M_1} (r_i - \hat{r}_i)^2 \quad (13)$$

where $\hat{r}_i$ and r_i are the original and predicted inference strength values, respectively, and M_1 is the number of explicit inference relations.

Algorithm 1 LMM training algorithm based on batch gradient descent

Input: Strengths of implicit inference relations $\hat{\mathbf{r}}$, Corresponding correlation vectors $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_{M_1}]$

Parameter: Learning rate $\eta = 0.01$, Maximum iterations $T_{max} = 1000$, Convergence threshold $\tau = 10^{-6}$

Output: ω, μ

```

1: Initialize parameters:  $\omega \leftarrow$  random vector,  $\mu \leftarrow 0$ .
2: for  $T$  from 1 to  $T_{max}$  do
3:   Predict inference strengths:  $\mathbf{r} \leftarrow \mathbf{C}\omega + \mu$ 
4:   Compute loss:  $L_{lmm}^T \leftarrow \frac{1}{M_1} \sum_{i=1}^{M_1} (\hat{r}_i - r_i)^2$ 
5:   if  $|L_{lmm}^{T-1} - L_{lmm}^T| < \tau$  then
6:     break
7:   end if
8:    $g_\omega \leftarrow -\frac{2}{M_1} \mathbf{C}^T (\hat{\mathbf{r}} - \mathbf{r})$ 
9:    $g_\mu \leftarrow -\frac{2}{M_1} \sum_{i=1}^{M_1} (\hat{r}_i - r_i)$ 
10:   $\omega \leftarrow \omega - \eta \cdot g_\omega$ 
11:   $\mu \leftarrow \mu - \eta \cdot g_\mu$ 
12: end for
13: return  $\omega, \mu$ 

```

The LMM training algorithm is shown in Algorithm 1. After LMM training, the strengths of implicit inference relations can be predicted based on the corresponding correlation vectors. To further determine whether an implicit inference relation between two data fields is reasonable, we identify two pairs of the most non-inference-related data fields: the maximum bus voltage magnitude and the minimum reactive power output limit of the generator, and the minimum bus voltage magnitude and the maximum reactive power output limit of the generator, to use their predicted inference strength values as a threshold. Only inference relations in which the strengths are higher than this threshold will be reserved.

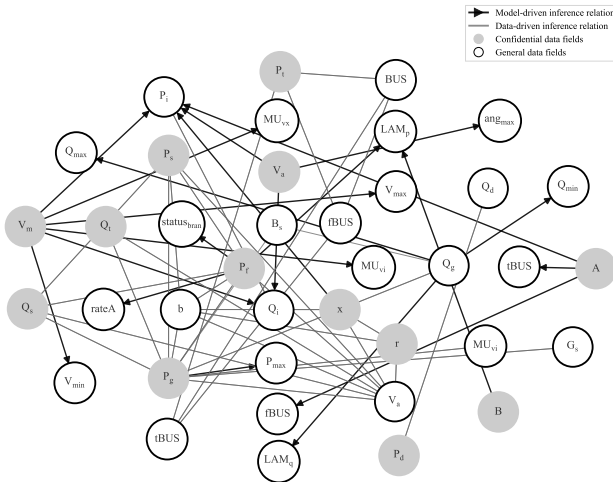


Fig. 2. The complete inference relation graph of the target data fields.

To intuitively visualize the complexity of DIL risks, we con-

struct the complete inference relation graph based on explicit and implicit inference relations, as illustrated in Fig. 2. The data correlations are calculated based on the dataset generated on the IEEE 30-bus system (as introduced in Section VI). This graph maps the intricate coupling relationships between different data types within the CPPG. In this visualization, black hollow circles denote general data fields, while gray solid circles represent confidential data fields which are the potential targets of inference attacks. The lines connecting these nodes represent the identified inference pathways. Specifically, black directed edges indicate model-driven relations, where the arrow points from the inferred target to the source, which is consistent to the direction of risk propagation. In contrast, gray undirected edges represent data-driven relations based on implicit statistical correlations mined from historical data. The dense connectivity observed in the figure demonstrates that confidential parameters are often coupled with multiple general data fields, confirming that risk can easily propagate through these inference pathways.

C. The IGNN Method for Assessment of Risk Propagation by DIL

To directly address the challenges of DIL in CPPG, where risk propagates dynamically through complex physical and statistical couplings, in this section we propose an IGNN method. The sensitivity assessment is framed as a node embedding task specifically designed to model the risk propagation by DIL. Unlike generic graph models, the proposed IGNN is tailored to capture the causal directionality and intensity of DIL pathways. As shown in Fig. 1, the overall model employs a multi-layer attention network to progressively fuse local neighborhood attributes and correlations, thereby transforming static data features into a global estimate of data sensitivity that fully accounts for dynamic data leakage risks.

1) *Node Feature Design*: The initial feature vector for each node consists of several components.

- Initial Sensitivity Value: Calibrated by the AHP-based model in Section V-A.
- Data Type: Variable (such as voltage phasor) or constant (such as transmission constraint).
- Topological Features: Include the number of connected edges and the average edge weight, reflecting the global importance of the node and the strength of its association with other nodes.

2) *Incorporation of Inference Matrix*: Compared to the data scenario of GAT that only considers whether nodes are connected, the strength and directionality of the connections also need to be considered in our inference relation graph. Therefore, we propose an inference-weighted GAT that integrates the directionality and strength of the inference relations into the learning process. The core of this method lies in the design of an *inference matrix* $\mathbf{R}$, which represents the structure of the inference relation graph. The elements $\mathbf{R}_{ij}$ are derived from the LMM calculation results as detailed in Section V-B3, which indicates the inference strength between node i and j . Moreover, $\mathbf{R}$ is designed to handle mixed graphs that contain directed and undirected edges:

- A directed edge from node i to j (which indicates an explicit inference relation from the data field i to j) is represented by a non-zero weight $\mathbf{R}_{ij}$, while the reverse path $\mathbf{R}_{ji}$ is set to zero. This asymmetry encodes the causal directionality of the inference.
- An undirected edge (which indicates an implicit inference relation between the data field i and j) is represented by symmetric weights, where $\mathbf{R}_{ij} = \mathbf{R}_{ji}$.

We integrate the inference matrix as a soft constraint within the GAT's aggregation mechanism. The inference strength $\mathbf{R}_{ij}$ modulates the impact of a neighboring node j on the target node i . The feature aggregation step is refined based on Eq. 4 and is shown as follows:

$$\mathbf{h}_i = \sigma \left[\sum_{j \in \mathcal{N}(i)} \alpha_{ij} (\mathbf{R}_{ij} \cdot \mathbf{W} \mathbf{h}_j) \right] \quad (14)$$

where σ denotes an activation function (e.g. ELU), and the weight $\mathbf{R}_{ij}$ scales the impact of the transformed features of node j on node i . Eq. 14 allows the model to learn not only the connectivity of the nodes but also the directionality and strength of the connections, thus constructing a more accurate feature propagation mechanism.

Because the data fields in CPPG exhibit complex, non-Euclidean coupling relationships that represent a graph structure, a GNN approach is necessary rather than CNN or RNN. It is also important to clarify that this design addresses specific challenges in DIL assessment that standard weighted or directed GNNs cannot effectively resolve. Existing directed GNNs typically treat all edges as uniform channels or learn weights based solely on feature similarity. However, in DIL, high feature similarity between data nodes does not necessarily imply a high inference risk (e.g., two unrelated physical parameters may be numerically close but physically uncoupled). By comparison, the self-attention mechanism in GAT allows the model to dynamically learn and assign distinct importance coefficients to different neighbors, effectively highlighting the most critical inference pathways. This capability is essential for accurately quantifying the accumulation of sensitivity. On this basis, the proposed IGNN distinguishes itself by imposing a physics-based prior via the inference matrix $\mathbf{R}$. Unlike standard GAT where attention mechanisms are purely data-driven, IGNN forces the feature aggregation to respect the physical inference causality derived from the LMM model. This ensures that the risk propagation is modulated by the actual inferability of the data rather than coincidental statistical similarities, a capability not realizable by generic weighted GAT variants.

3) *The Architecture Design of IGNN*: The architecture of the proposed IGNN consists of three layers, designed to balance model complexity and generalization. In our method, the first two layers use multi-head attention to capture local and global features, while the third layer utilizes a single-head attention mechanism for final sensitivity estimation. This architecture ensures effective feature aggregation and refinement while avoiding over-smoothing or overfitting.

The first layer is to aggregate the neighborhood features to quantify the impact of risk propagation by data inference

relations. This layer uses a multi-head attention mechanism with $K = 4$ heads:

- For the k -th head, compute the attention score:

$$e_{ij}^{(k)} = \text{LeakyReLU} \left(\mathbf{a}^{(k)T} [\mathbf{W}^{(k)} \mathbf{h}_i \parallel \mathbf{W}^{(k)} \mathbf{h}_j] \right) \quad (15)$$

- Since variable data cannot be inferred from constant data, we construct a mask matrix $\mathbf{M}$ to block information exchange by unreasonable inference paths.

$$M_{ij} = \begin{cases} 0, & \text{if } i \text{ is constant and } j \text{ is variable} \\ 1, & \text{otherwise} \end{cases} \quad (16)$$

After Applying the mask:

$$e_{ij}^{(k), \text{mask}} = e_{ij}^{(k)} \times M_{ij} \quad (17)$$

then normalize to obtain:

$$\alpha_{ij}^{(k)} = \frac{\exp(e_{ij}^{(k), \text{mask}})}{\sum_{j' \in \mathcal{N}(i)} \exp(e_{ij'}^{(k), \text{mask}})} \quad (18)$$

- By applying Eq. 14, features can be aggregated for each head

$$\mathbf{h}_i^{(k)} = \text{LeakyReLU} \left[\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} (\mathbf{R}_{ij} \cdot \mathbf{W}^{(k)} \mathbf{h}_j) \right] \quad (19)$$

- Then concatenate the outputs from all heads

$$\mathbf{h}_i' = \parallel_{k=1}^4 \mathbf{h}_i^{(k)} \quad (20)$$

The second layer utilizes the same multi-head structure as the first layer, to further refine the aggregated features:

$$\mathbf{h}_i'' = \parallel_{k=1}^4 \mathbf{h}_i'^{(k)} \quad (21)$$

The third layer employs a single-head attention mechanism to obtain the final assessment of data sensitivity. The attention score computation and normalization are the same as in Eq. 15 to Eq. 19. And we can get the final prediction of sensitivity for data field i in the aggregation stage as:

$$\hat{s}_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^3 (\mathbf{R}_{ij} \cdot \mathbf{W}^3 \mathbf{h}_j'') \quad (22)$$

4) *The Configuration and Training of IGNN*: The training objective function is formulated by the Mean Squared Error (MSE) loss, which compares the predicted sensitivity $\hat{s}_i$ with the pre-calibrated sensitivity s_i :

$$\mathcal{L} = \frac{1}{N_1} \sum_{i=1}^{N_1} (\hat{s}_i - s_i)^2$$

where N_1 is the number of confidential data fields whose sensitivity is definite.

To ensure optimal performance, we adopted a sequential tuning strategy to investigate the parameter configuration: first, we identified the optimal learning rate based on validation stability and then determined the sufficient number of epochs under the selected rate. The validation loss refers to the mean squared error (MSE) evaluated on a held-out validation set, which reflects the model's generalization ability rather than its fitting error on the training data.

The sensitivity of the model to the learning rate is investigated by training IGNN with $\eta \in \{0.01, 0.005, 0.001, 0.0005\}$ and recording the validation MSE for up to 1000 epochs. The learning curves in Fig. 3 indicate that a high learning rate, such as $\eta = 0.01$, leads to unstable optimization. The validation loss decreases sharply initially but soon diverges monotonically, indicating severe oscillations and overfitting. The intermediate value $\eta = 0.005$ shows faster initial convergence, but exhibits an upward trend after several tens of epochs, suggesting that it is still too aggressive for sustained training. In contrast, lower rates ($\eta = 0.001$ and $\eta = 0.0005$) produce more stable curves. Although $\eta = 0.0005$ continues to improve slowly, it requires a significantly larger computational budget. Consequently, $\eta = 0.001$ provides the best compromise between convergence speed, stability, and final accuracy, and is therefore adopted as the default learning rate.

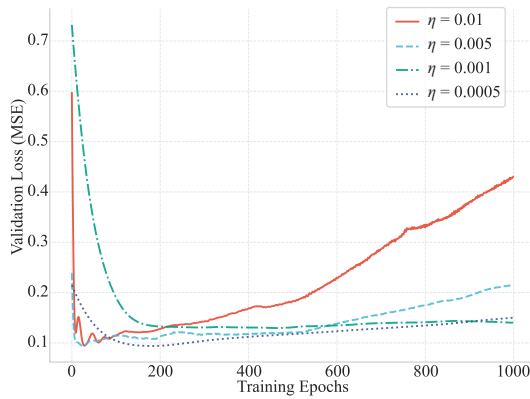


Fig. 3. Sensitivity of IGNN to the learning rate η .

After that, with η fixed at 0.001, the training dynamics is further examined to determine the appropriate T_{max} . Fig. 4 reports the evolution of the training loss and validation loss (both in MSE) under the selected learning rate. Since the proposed model integrates complex physics-based inference matrices, monitoring the validation curve ensures that the optimization process does not result in overfitting to the specific training snapshots. Both losses drop rapidly during the first 150 to 200 epochs, indicating an effective optimization phase. Beyond 200 epochs, the losses flatten out with only minor fluctuations, suggesting that the model has essentially converged and further training yields negligible gains while increasing the risk of mild overfitting. Based on these observations, the maximum number of epochs is set to $T_{max} = 300$ for all subsequent experiments, which safely covers the convergence region while ensuring computational efficiency.

VI. NUMERICAL EXPERIMENTS AND ANALYSES

A. Experiment Setup

To test our data sensitivity assessment method, 40 key data fields are consolidated as the baseline dataset (from line 1 to line 40 in the nomenclature in Section I), referring to the universality and importance of these data. The specific values of the target data fields are obtained by IEEE 30-, 57-bus systems, the Illinois ACTIV 200-bus system [42], and

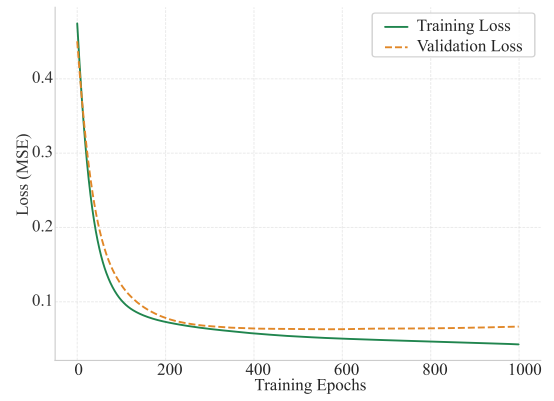


Fig. 4. Training dynamics of IGNN with $\eta = 0.001$.

the Case 1354-pegase system. Using MATPOWER (a power system simulation software on MATLAB), 100 snapshots of data are generated for each target data field on 30, 57, 200, and 1354 bus systems. Active and reactive loads at various buses and the power output of generators are adjusted to simulate various operating states to get multiple snapshots of data. The experiments are structured as follows:

- 1) A comparison of the results of the data sensitivity assessment is presented on the 30, 57, 200, and 1354 bus systems to evaluate the consistency of the assessment on different CPPGs and also reveal the differences.
- 2) The reliability of the data sensitivity assessment is verified by using identified high-risk data (with high sensitivity) to launch DIL attacks and testing the accuracy of inference results.
- 3) The proposed method is further compared with other risk assessment methods, and ablation studies are performed to verify the functionality of the modules designed for our framework.

B. Cross-System Comparison and Analysis

The effectiveness of the proposed data sensitivity assessment method is evaluated on IEEE 30-, 57-bus systems, the ACTIV 200-bus system, and the 1354-pegase system. After calibration of the initial sensitivity of the target data fields, we calculate the final sensitivity 10 times based on the multiple snapshots of CPPG data in different operating states, and the aggregated results of data sensitivity are presented in Fig. 5, Fig. 6, Fig. 7, and Fig. 8. Specifically, data fields with higher sensitivity scores not only indicate significant data security risks, more importantly, exhibit strong inference couplings with confidential data. Consequently, the direct release of such data implies a higher probability of DIL, acting as a leakage gateway for attackers to infer inaccessible critical data. Therefore, these quantitative assessment results provide solid basis for power system to implement targeted protection strategies, such as stricter access control policies, data perturbation or differential privacy techniques, thereby mitigating the potential risk of information leakage.

According to the results, the interquartile ranges are relatively small for most data fields across all four systems,

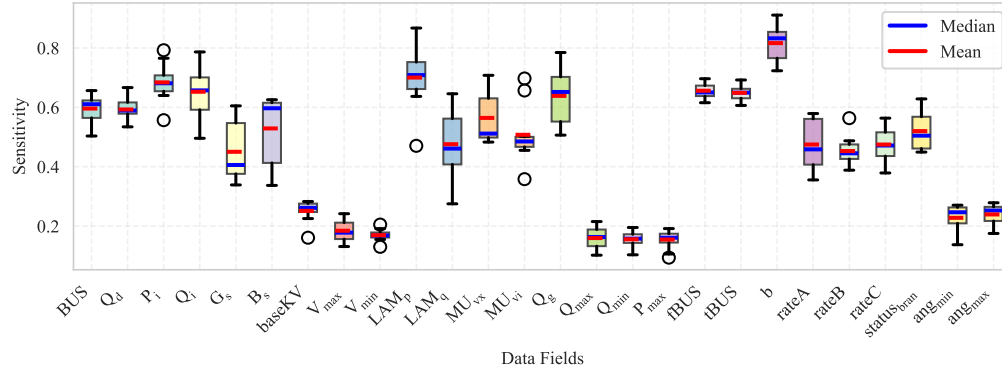


Fig. 5. Final sensitivities of target data fields in the IEEE 30-bus system. Each boxplot includes 10 assessment results under distinct system states.

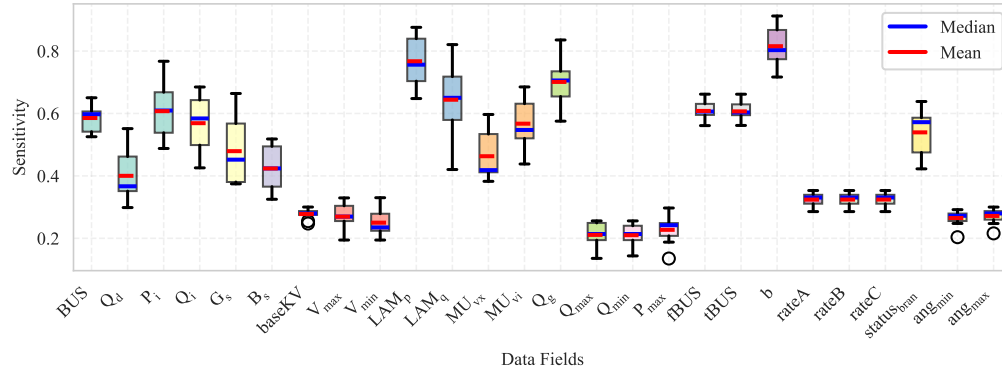


Fig. 6. Final sensitivities of target data fields in the IEEE 57-bus system. Each boxplot includes 10 assessment results under distinct system states.

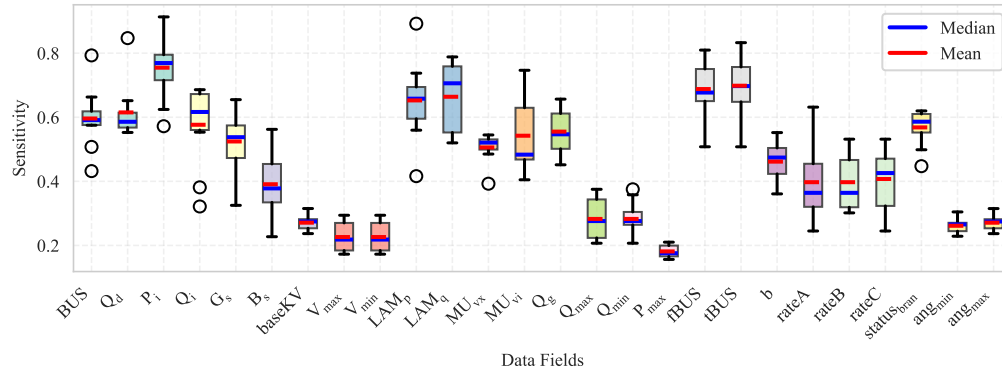


Fig. 7. Final sensitivities of target data fields in the ACTIVS 200-bus system. Each boxplot includes 10 assessment results under distinct system states.

indicating that the assessment of data sensitivity is stable and will not be significantly interfered with by changes in system states. Moreover, the results across the systems also reveal that, while some data fields are universally high-sensitivity, others depend on the specific system properties.

For instance, fields highly related to the grid topology, such as $fbus$, $tbus$, and b , consistently exhibit high sensitivities and low deviations across the three systems. This indicates their fundamental importance in data inference pathways, as they define the physical topology and parameters of the power grid. It is worth noting that although $status_{bran}$ reflects the

operational state of the topology (i.e., whether a branch is in-service), its assessed sensitivity is relatively lower than $fBUS$, $tBUS$, and b . This may be because $status_{bran}$ is a binary variable (0 or 1) that essentially acts as a switch, providing limited information about the specific grid structure. $fBUS$, $tBUS$ and b explicitly define the global topology and line parameters of power grid, consequently resulting in higher sensitivity scores.

Moreover, some other fields have various sensitivities in different power grids. For example, comparing Fig. 5 and Fig. 7, the sensitivity of b is notably high in the 30-bus system

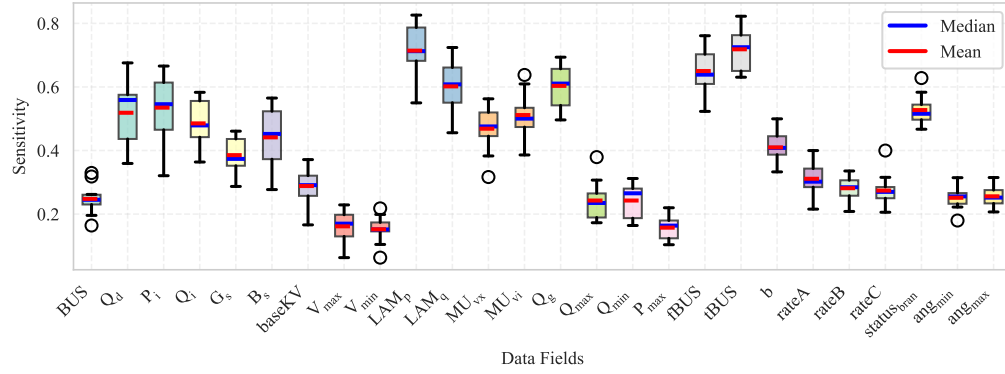


Fig. 8. Final sensitivities of target data fields in the Case 1354-pegase system. Each boxplot includes 10 assessment results under distinct system states.

and the 57-bus system, but significantly lower in the 200-bus system. Reactive power prices LAM_q are identified as highly sensitive in the 57-bus system and the 200-bus system, but not in the 30-bus system. These divergences demonstrate that the IGNN framework successfully captures different inference pathways and associated risks due to unique system properties and data interdependencies in different power systems.

To further valid the scalability of the proposed method on large-scale power grids with thousands of nodes and complex data structures, we applied the assessment framework to the Case 1354-pegase system. The assessment results in Fig. 8 show that the sensitivity distribution in the 1354-bus system maintains a similar regularity. Topology-related parameters exhibit consistently high sensitivity, while the sensitivity of operational variables fluctuates reasonably with different system states. This demonstrates that our method can effectively handle the complexity of large-scale grids and maintain stable assessment performance.

C. The Assessment Validation based on DIL Test

A vital objective of our assessment method is to quantify the potential DIL risks hidden in general CPPG data. To verify the reliability of the proposed sensitivity metric, we conduct a DIL simulation where a neural network model serves as an adversary. In this context, the DIL attack is defined as the process where the adversary utilizes accessible general data to infer confidential target data. Consequently, the inference accuracy of the model reflects the feasibility and severity of the DIL attack, which means a higher inference accuracy implies a higher risk of data leakage.

To verify the assessment results, the general data are classified into three risk levels based on the sensitivity scores s calculated by our IGNN framework: low sensitivity ($0 \leq s < 0.35$), medium sensitivity ($0.35 \leq s < 0.65$), and high sensitivity ($0.65 \leq s \leq 1.0$). Then, the general data at different sensitivity levels could be used to infer the confidential data, to verify whether the results of the sensitivity assessment can truly reflect the DIL risk level. The mean percentage accuracy (MPA) of the inferred confidential data is developed to evaluate DIL attacks, which is calculated by the following

equation:

$$MPA = 1 - \frac{1}{N_y} \left\| \frac{\hat{\mathbf{y}} - \mathbf{y} + \mathbf{1}}{|\mathbf{y}| + \mathbf{1}} \right\|_1 \quad (23)$$

where $\hat{\mathbf{y}}$ are the inferred values of the confidential data $\mathbf{y}$ and N_y is the dimension of $\mathbf{y}$.

To rigorously evaluate the robustness of the assessment method, we further analyze the impact of both the attacker's model capacity and the amount of available training data. We assume a realistic scenario where the attacker trains the inference model using a dataset of historically leaked sensitive data as labels. This simulates a situation where an adversary has obtained partial system operations (e.g., via a previous breach or insider threat) and seeks to train a model to infer future confidential data based on real-time general data. We construct three types of adversaries to investigate these factors: 1) *Mediocre adversary (limited data)*: Uses a simple neural network consisting of an input layer, two hidden layers with ReLU activation, and an output layer, trained on a limited historical dataset (20% of total samples), serving as a baseline to verify if high-sensitivity data remains vulnerable even with minimal attacker resources. 2) *Mediocre adversary (abundant data)*: Uses the same simple network but trained on an abundant historical dataset (50% of total samples), designed to analyze how the volume of training data affects inference performance. 3) *Powerful adversary (abundant data)*: Uses a Deep Neural Network (DNN) with five hidden layers and optimized hyperparameters, trained on the abundant dataset, representing a sophisticated attacker with maximum capabilities.

The reason for the selection of a mediocre adversary is that if confidential information can be inferred with high accuracy even by a simplistic model, it validates that the source data possesses a strong intrinsic correlation with the targets, thereby confirming the high sensitivity assessed by our method. This simulation prevents the assessment from being overly sensitive by ensuring that data labeled as high-risk are vulnerable even to basic attacks. Thus, different risk-level general data are used as the input of the mediocre adversary model to test its inference accuracy on the IEEE 30-bus system, while several power flow data and line parameter (widely recognized as

confidential data in CPPG) are chosen as the target of the inference attack.

As shown in Table VII, the inference accuracy exhibits a significant positive correlation with the risk levels. The high risk group yields the highest inference accuracy across all confidential parameters, whereas the low risk group provides weak inference capabilities. This contrast empirically validates that our IGNN-based assessment successfully distinguishes data that implies real DIL threats from safe data, providing a reliable basis for hierarchical data protection.

TABLE VII
COMPARISON OF MPAS OF DIL USING DATA WITH DIFFERENT RISK LEVELS ON IEEE 30-BUS SYSTEM

Confidential Data	Low Risk ($s < 0.35$)	Medium Risk ($0.35 \leq s < 0.65$)	High Risk ($s \geq 0.65$)
x	0.5007	0.5386	0.9145
P_f	0.3461	0.4258	0.8704
P_t	0.1220	0.1318	0.3399
Q_f	0.3352	0.3864	0.8571
Q_t	0.2418	0.3073	0.6607
P_s	0.1129	0.1095	0.5378

TABLE VIII
MPAS OF INFERRED CONFIDENTIAL DATA IN 30-, 57-, 200-, AND 1354-BUS SYSTEMS

Confidential data	30-bus	57-bus	200-bus	1354-bus
x	0.9145	0.9209	0.6465	0.3231
P_f	0.8704	0.6851	0.4268	0.1556
P_t	0.3399	0.4293	0.6295	0.1894
Q_f	0.8571	0.7852	0.6522	0.2383
Q_t	0.6607	0.2493	0.3476	0.0956
P_s	0.5378	0.1485	0.4327	0.2173

* The MPAs beyond 0.6 are in bold, which indicate a threatening inference of the confidential data.

Table VIII shows the MPAs of DIL attacks on the 30-, 57-, 200-, and 1354-bus systems. According to the results, the confidential data fields exhibit varying degrees of DIL risks due to the identified risk data. For example, the MPAs of branch reactance x and active power flow P_f reach 0.9145 and 0.8704 in the 30-bus system, which are significantly accurate for blind inferences. Similarly, high inference risks are observed in the 57-bus system, where x and reactive power flow Q_f achieve MPAs of 0.9209 and 0.7852, respectively. It can also be noticed that the MPA of P_s (active power loss) is relatively lower across most systems. This may be because the calculation of power loss involves more parameters and is difficult to infer blindly.

In addition, it can be seen that with the system scale increasing, the accuracy of DIL generally shows a downward trend. This is because the increase in system complexity and data dimensions raises the difficulty of inference calculation; consequently, the inference accuracy in the 1354-bus system drops below 0.4. Nevertheless, this does not indicate the non-existence of DIL risks in large-scale systems. For example, the MPAs of x , P_t , and Q_f still exceed **0.6** in the 200-bus system. Therefore, the experiments show that the identification of DIL risk data based on data sensitivity assessment is quite significant for CPPG.

Then, by using the high risk data as input, the inference accuracy of these three adversary models is compared on the IEEE 30-bus system, and the results are presented in Table IX. According to the results, both the amount of training data and the model complexity positively correlate with inference accuracy. The mediocre adversary with abundant historical data obtains higher MPA (e.g., x improves from 0.8496 to 0.9145), confirming that the accumulation of historical leaks exacerbates DIL risks. Furthermore, the powerful adversary achieves high accuracy across all fields (e.g., 0.6981 for Q_t), demonstrating the upper bound of the risk. However, crucially, even the *Mediocre Adversary with Limited Data* achieves high accuracy (over 0.8) on key parameters like x and P_f . This result confirms that the data identified as high-sensitivity by our method exposes the system to severe risks even when the attacker has limited resources and data, thereby validating the robustness and necessity of the proposed sensitivity metric.

TABLE IX
COMPARISON OF MPAS UNDER DIFFERENT ADVERSARY CAPABILITIES AND DATA VOLUMES ON 30-BUS SYSTEM

Confidential Data	Mediocre 1	Mediocre 2	Powerful
x	0.8496	0.9145	0.8774
P_f	0.8392	0.8704	0.8915
P_t	0.2157	0.3399	0.5124
Q_f	0.6424	0.8571	0.8205
Q_t	0.4285	0.6607	0.6981
P_s	0.4016	0.4897	0.5378

D. Comparative Analysis and Ablation Studies

In this section, we conduct a comprehensive evaluation to verify the superiority and internal rationality of the proposed IGNN framework. IGNN is compared with the state-of-the-art rule-based deep learning (DL) method [6] and the GCN-based method [13] to demonstrate its performance advantages in effectively quantifying data sensitivity and identifying DIL risks. Then, the ablation study is performed to validate the necessity of the hybrid mechanism that combines the data-driven LMM with the rule-based AHP. Finally, a robustness analysis is conducted to verify that the model's assessment capability relies more on the intrinsic inference structure rather than being dominated by initialization priors.

1) *Comparison with other assessment methods*: By comparison, the GCN-based method relies solely on topological connectivity without considering the heterogeneous edge weights derived from data correlations, while the rule-based method relies solely on static expert rules with the enhancing of Bayesian network to perform an integrated safety and security risk analysis. We first analyze the distribution of the results of the sensitivity assessment.

To validate the accuracy of the risk identification, we utilized the high-sensitivity data identified by each method to perform DIL attacks. The MPA of the confidential data inferred by these methods is presented in Table X. The IGNN method achieves better inference accuracy in all confidential data fields (e.g., 0.9145 for x), confirming its superior capability in locating the most vulnerable data sources. In contrast,

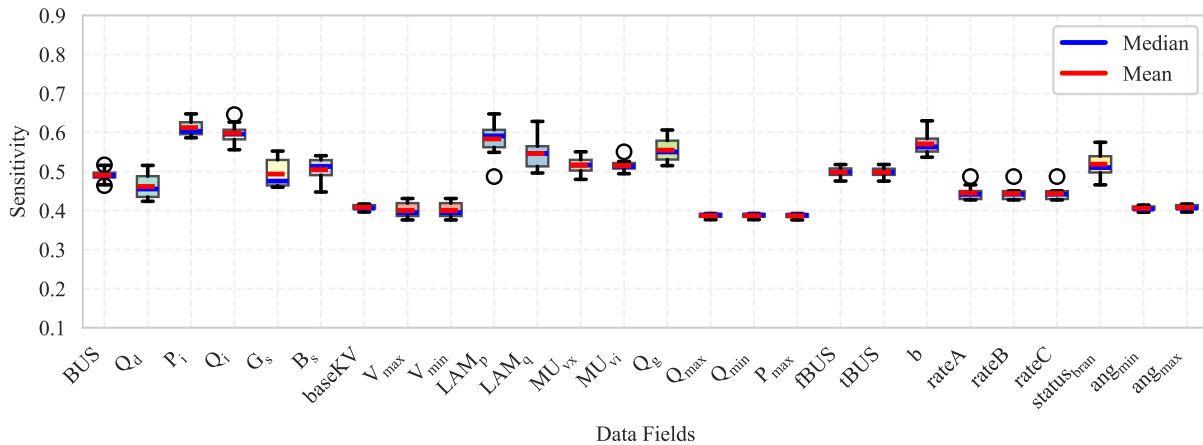


Fig. 9. The assessment results of our method without the AHP evaluation module.

the lower MPAs of the GCN-based and rule-based methods suggest that they miss key inference pathways, leading to an underestimation of potential leakage risks.

TABLE X
COMPARISON OF DIL INFERENCE ACCURACY (MPA) BASED ON
DIFFERENT ASSESSMENT METHODS

Method	x	P_f	P_t	Q_f	Q_t	P_s
IGNN	0.9145	0.8704	0.3399	0.8571	0.6607	0.4897
Rule-DL	0.6150	0.1250	0.1251	0.5117	0.2574	0.1691
GCN	0.7044	0.6476	0.2647	0.8027	0.4267	0.3214

2) *Ablation Studies*: The core of the IGNN framework lies in the integration of prior knowledge (AHP) and dynamic correlation learning (LMM) into the GNN framework. Therefore, to verify the contribution of each component, the AHP module is first removed and a uniform initialization strategy (confidential data set to 1, all general data set to 0.5) is used. The results of the sensitivity assessment are shown in Fig. 9. The results indicate that without the baseline provided by AHP, the model struggles to converge to a distinguishable state. The final sensitivity scores for most data fields do not distinguish high-risk data fields significantly from low-risk ones. This confirms that AHP provides a necessary baseline of intrinsic security risks, preventing the model from getting stuck in local optima during feature aggregation.

Then, the LMM-based inference matrix is removed, and the graph is constructed using only explicit physical connections, ignoring implicit data-driven correlations. The results in Fig. 10 show that the method can still obtain similar results, but the assessment results show large discrimination and minimal variance (narrow boxplots), implying that the model becomes insensitive to changes in the operating states of the system. This suggests that the LMM module is also important in capturing the dynamic nature of DIL risks.

In summary, the absence of either component leads to a compromised assessment framework that is either indistinguishable or unresponsive to real-world data variations.

3) *Robustness testing of the proposed method for initial sensitivity calibration*: To further validate the robustness of the

IGNN method and ensure that it is not simply fitting the initial calibration results of the data sensitivity, we intentionally introduced a strong bias by initializing the sensitivity of all confidential data to 1 and all general data to a high sensitivity of 0.9. This simulates an extreme scenario where the system initially assumes that all data are highly sensitive.

The results of sensitivity differentiation are illustrated in Fig. 11. Despite uniformly high sensitivity initialization, the IGNN method still obtains discriminative sensitivity assessment results of different data fields after the training stage. It effectively suppressed the sensitivity of non-critical parameters (lowering them from 0.9 to appropriate lower levels) while maintaining high scores for true high-risk data. This shows that the proposed method has strong convergence stationarity. It shows that the final sensitivity assessment is driven more by the intrinsic inference structure learned from the graph, rather than being dominated by the subjective initial priors.

E. Computational Efficiency and Resource Utilization

To assess the practical deployability of the proposed IGNN in Cyber-Physical Power Grid (CPPG) settings, we benchmark its computational efficiency and resource usage against standard GCN and standard GAT. All models are trained and evaluated on the same hardware platform (NVIDIA GeForce RTX 4060 GPU, 16GB RAM) under an identical software stack (PyTorch 2.1, CUDA 12.1). Table XI reports the average training time per epoch, total training time, peak GPU memory footprint, and online inference latency.

Overall, IGNN introduces a modest computational overhead (0.0194 s/epoch) relative to standard baselines. This cost mainly arises from incorporating the inference matrix $\mathbf{R}$ and the mask-modulated multi-head attention $\mathbf{M}$, which explicitly enforce the physical and statistical coupling relations rather than uniform neighborhood aggregation in conventional GNNs. Despite the additional computations, the absolute runtime of IGNN remains low: total training completes within 2 seconds and online inference takes 6.9 ms. Meanwhile, IGNN's peak GPU memory usage (16.89 MB) is comparable

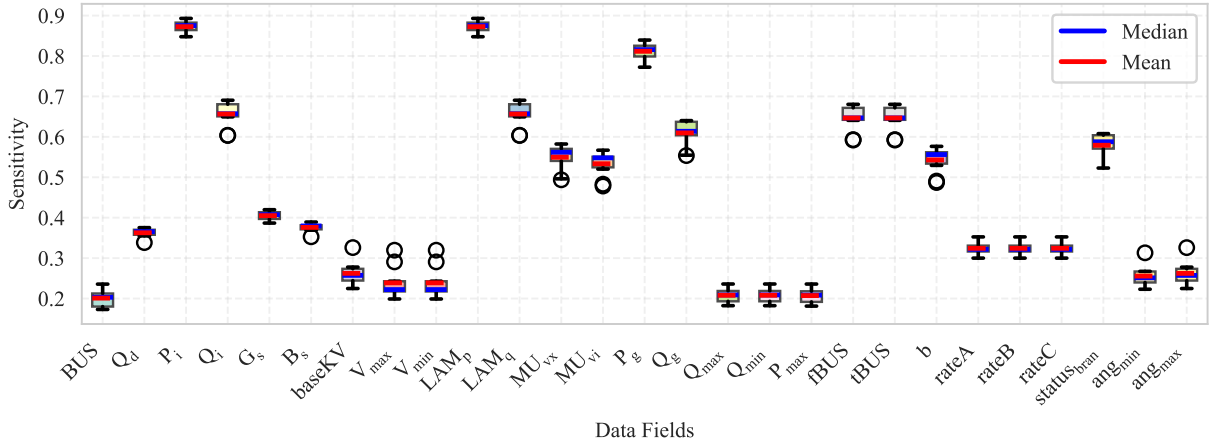


Fig. 10. The assessment results of our method without the LMM module.

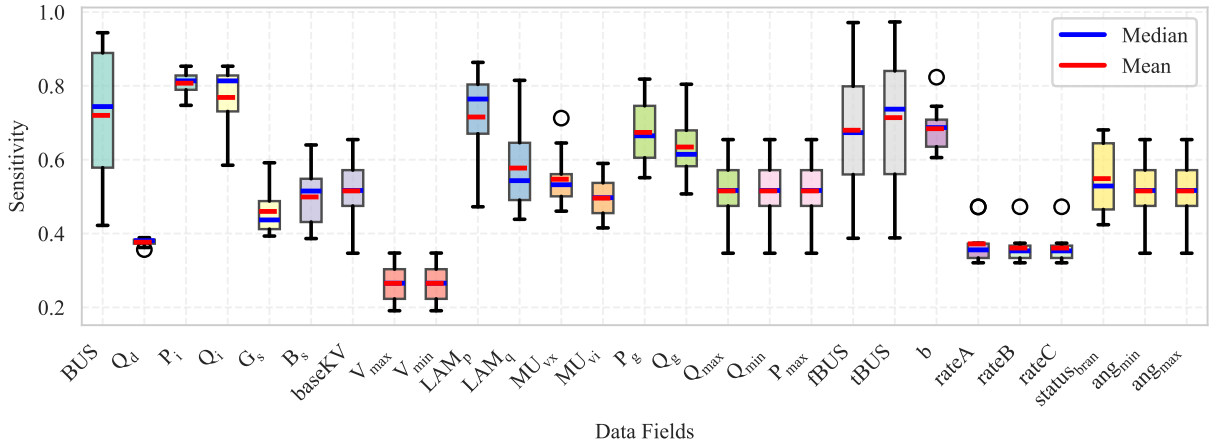


Fig. 11. The assessment results with strong bias introduced in initial sensitivity calibration on 30-bus system.

to the baselines, indicating that the enhanced inference logic does not incur excessive memory overhead.

TABLE XI
COMPUTATIONAL EFFICIENCY AND RESOURCE UTILIZATION

Method	Time/epo (s)	Train (s)	GPU Mem (MB)	Infer (s)
GCN	0.00606	0.606	16.43	0.0031
GAT	0.01262	1.262	17.01	0.0058
IGNN	0.01940	1.940	16.89	0.0069

For assessments of large-scale power datasets, the efficiency of IGNN can be further improved by graph sparsification, such as increasing the LMM threshold to prune weak inference edges and reduce graph density. Moreover, some model quantization methods can also be utilized, e.g., convert FP32 to INT8 to accelerate inference on resource-constrained hardware.

VII. DISCUSSION

While this study primarily validates the proposed framework using transmission grid data, the underlying logic of the IGNN-based sensitivity assessment is generic and scalable

to a broader range of CPPG data scenarios. The framework can be effectively extended to domains where physical or statistical coupling relationships can be modeled as graph edges. For instance, with the proliferation of Advanced Metering Infrastructure (AMI) in distribution networks, fine-grained user consumption data is collected, which contain implicit correlations with user behaviors. Then our framework can model users and appliances as nodes to capture the risk of privacy inference, such as inferring household occupancy from load curves. Similarly, in scenarios involving high penetration of Distributed Energy Resources (DERs), generation data is strongly coupled with meteorological information. The proposed method can assess the risk of inferring critical grid locations or dispatch strategies from open-source weather data.

Beyond broadening the scope of assessment, the future direction of our work is to guide the implementation of data protection measures. Based on the quantitative sensitivity assessment, a hierarchical data protection framework can be established to balance data utility and security. For data identified with high DIL risk, a balance between utility and privacy is required. Techniques such as Differential Privacy (DP) or data masking can be applied, where the noise level

or masking intensity is dynamically adjusted based on the specific sensitivity score to disrupt inference pathways while maintaining data usability. Conversely, for data assessed as low risk, general monitoring is sufficient, allowing data to be shared with authorized third parties to facilitate business operations with minimal restrictions. By integrating the proposed assessment method as the policy engine, the protection framework can dynamically adjust defense strategies, ensuring that security resources are precisely allocated to the most vulnerable data nodes.

Moreover, considering the rapid advancement of Large Language Models (LLMs), their potential integration into DIL assessment presents a promising avenue for future research. LLMs could enhance the proposed IGNN framework by processing unstructured data (e.g., operation logs and maintenance manuals) to extract semantic inference pathways that complement model-driven and data-driven correlations. However, challenges such as lack of prior knowledge and hallucinations may hinder the application of LLMs to extract reliable inference pathways. Therefore, the supervision based on expert knowledge may still be important in the assessment process.

VIII. CONCLUSION

In this paper, we establish a quantitative framework for assessing CPPG data sensitivity under DIL threats based on a novel IGNN. Unlike traditional methods that treat data fields in isolation, we redefine data sensitivity as a dynamic metric composed of intrinsic risk and propagated risk. By mapping heterogeneous model-driven and data-driven couplings into an inference relation graph, the proposed inference-weighted IGNN effectively captures the directional flow of leakage risks. Validations on multiple benchmark systems, up to the Case 1354-pegase system, confirm the framework's reliability in identifying data that serves as inference bridges.

In future work, we aim to extend the assessment range to include distribution networks and market data, and to integrate these sensitivity metrics into a hierarchical data protection framework.

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